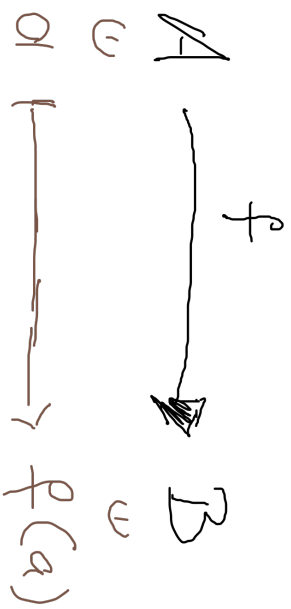


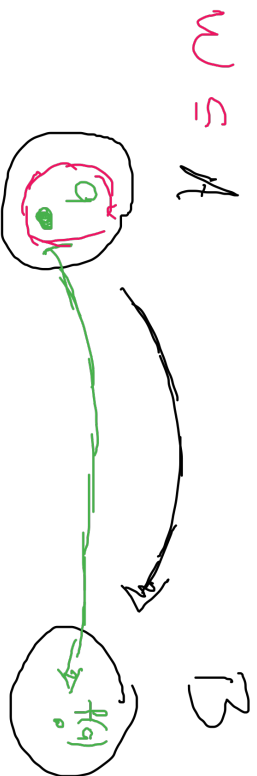
Week 9: functions, images, & surjectivity

Let  $A, B$  be sets.

A function is a "rule" that associates, to each  $a \in A$ , some  $b \in B$



Can't graph...



## Examples

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

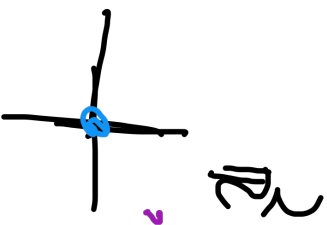
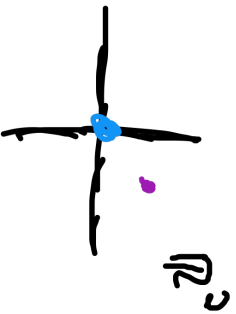
$$(x, y) \mapsto (x+y, x, y)$$

$$f(x, y) = (x+y, x, y)$$

$$f(0, 0) = (0, 0)$$

$$f(1, 1) = (2, 1)$$

can't graph



"unambiguous"  $\hat{=}$   $\forall a \in A, \exists$  exactly one output  $f(a) \in B$ .

"not multivalued"

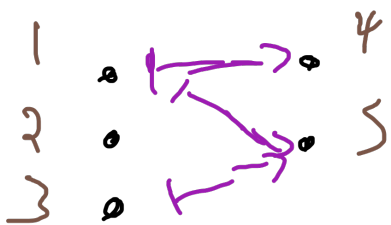
"passes the VLT" if  $A=B=\mathbb{R}$

Let go of the formulas

$$A = \{1, 2, 3\}, B = \{4, 5\}$$

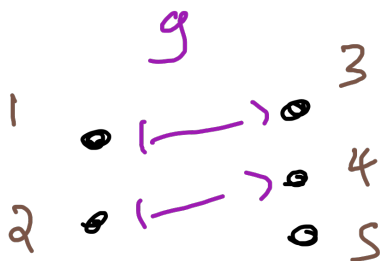


$$f(1) = 4 \quad f(2) = 4 \quad f(3) = 5$$



2 problems  
 Ambiguous (what is  $f(1)$ ?)  
 Didn't define  $f(2)$  ...

From far away ... just dots



5 is part of the codomain  
 5 is not an output

Domain

"  
inputs

codomain

potential outputs

range = image

Actual outputs

$A = \{x \mid x \text{ is a student in math 55}\}$

$B = \{\text{yes, no}\}$

$A \xrightarrow{f} B$

$f(x) = \text{Answer to "Is } x \text{ wearing glasses?"}$

$f(\text{Angela}) = \text{no}$

$f(\text{Jack}) = \text{no}$



$$\mathbb{Z} \xrightarrow{g} \mathbb{Z}$$

$$x \mapsto \begin{cases} x/2 & \text{if } x \in \mathbb{E} \\ 3x+1 & \text{o/w} \end{cases}$$

other wise

$$g(1) = 4$$

$$g(4) = 1$$

Caution

$$\mathbb{Z} \xrightarrow{h} \mathbb{Z}$$

$$x \mapsto x/2$$

invalid

b/c  $x/2 \notin \mathbb{Z}$

if  $x=1$

$$\mathbb{E} \rightarrow \mathbb{Z}$$

$$x \mapsto x/2$$

ok

# Indicator fcn of $G$

$$\mathbb{R} \xrightarrow{f} \mathbb{R}$$

$$x \mapsto \begin{cases} 0 & \text{if } x \notin \mathbb{Q} \\ 1 & \text{if } x \in \mathbb{Q} \end{cases}$$

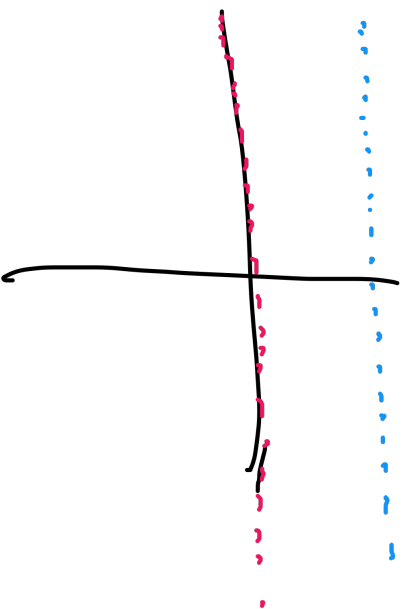
unambiguous,

$$g(0) = 1$$
$$g(22/7) = 1$$

$$g(\sqrt{2}) = 0$$
$$g(\pi) = 0$$

$$g(\tan 1) = ?$$

$$(x^5 + 7x + 1 = 0)$$



What does it mean for  $f = g$ ?

$$f: A \rightarrow B$$

$$g: C \rightarrow D$$

Defn: We say that  $f = g$  if

$$A = C, B = D, \text{ and } \forall a \in A, f(a) = g(a).$$

To show  $f \neq g$ , show  $A \neq C$ ,  $B \neq D$ , or

$$\exists a \in A \text{ s.t. } f(a) \neq g(a).$$

$\forall a, b \in A, f(a) = f(b)$  ↙ consistent  
if  $a = b, f(a) = f(b)$  ↙ uniqueness

Examples:

$$\mathbb{R} \xrightarrow{f} \mathbb{R}$$

$$x \mapsto 2x$$

$f \neq g$  b/c different domains  
and codomains.

$$\mathbb{Z} \xrightarrow{g} \mathbb{Z}$$

$$x \mapsto 2x$$

$g(\sqrt{2})$  is undefined w/c

$$\sqrt{2} \notin \mathbb{Z}$$

$$\mathbb{Z} \xrightarrow{h} \mathbb{E}$$

$$x \mapsto 2x$$

$$h \neq g$$

b/c different codomains.

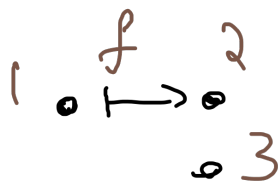
$$\mathbb{Z} \xrightarrow{h_2} \mathbb{E}$$

$$n \mapsto 2n$$

$$\forall b, h(b) = h_2(b)$$

$$\begin{matrix} \parallel & \parallel \\ 2b & 2b \end{matrix}$$

$$h = h_2$$



$$f \neq g$$

$$f(1) \neq g(1)$$

$$\begin{matrix} \parallel & \parallel \\ 2 & 3 \end{matrix}$$

$$f(n) = n+1$$

$$f(n) = 2n$$

inval

↓

$$S \mapsto S \cup I$$

$$P(B) = \{A \subseteq B\}$$

$$P(\mathbb{Z}) \longrightarrow P(\mathbb{Z})$$

$$A \in P(B) \iff A \subseteq B$$

$$S \xrightarrow{f} S \cup \{1\} = f(S)$$

$$\text{b/c } S \cup \{1\} \notin P(\mathbb{Z})$$

$$S \xrightarrow{g} S \cap E$$

$h$  is invalid

$$S \xrightarrow{h} S \cup \{\pi\}$$

$$f(E) = E \cup \{1\}$$

$$f(\{1, 3\}) = \{1, 3\} \cup \{1\} = \{1, 1, 3\}$$

$$f(\mathbb{Z}) = \mathbb{Z} \cup \{1\} = \mathbb{Z}$$

$$f(\emptyset) = \emptyset \cup \{1\} = \{1\}$$

$$g(\{1, 3\}) = \{1, 3\} \cap E = \{1\}$$

$$g(\mathbb{Z}) = \mathbb{Z} \cap E = E$$

$$\mathbb{R} \longrightarrow \mathcal{P}(\mathbb{R})$$

$$x \longmapsto (x, \infty)$$

$$(x, \infty) = \{a \in \mathbb{R} \text{ s.t. } x < a\}$$

$$x \longmapsto \{x\}$$

$$x \longrightarrow x$$

$$Y \longmapsto Y+1 \text{ invalid}$$

$$y \in Y+1 \notin \mathcal{P}(\mathbb{R})$$

# Common functions

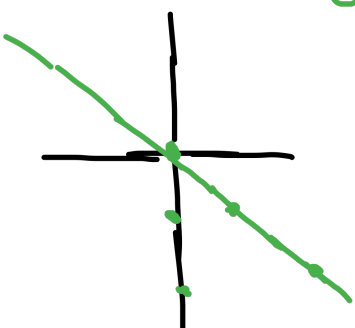
$A \xrightarrow{\text{id}_A} A$  "do nothing"

$$x \mapsto x$$

$$A = \mathbb{R}$$

$$\text{id}_A(x) = x$$

$$\text{id}(x) = x$$



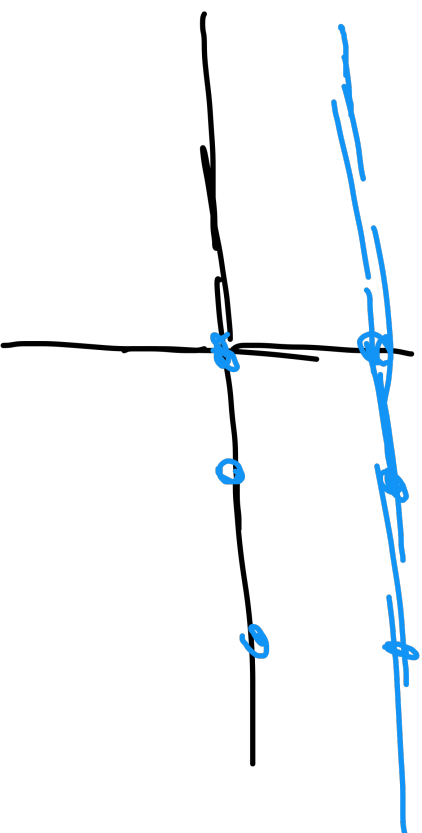
SpS  $B \neq \emptyset$  and let  $b \in B$ .

$$A \xrightarrow{c_b} B$$

$$A = B = \mathbb{R} \quad b = 1$$

$$a \mapsto b$$

$$c_b(a) = b$$



Defn: Let  $A$  and  $B$  be sets and  $f: A \rightarrow B$  be a function.

The image (range) of  $f$  is  
(write as  $\text{im } f$  or  $f(A)$ )

$$\text{im } f = \{ f(a) : a \in A \}$$

If  $W \subseteq A$ , define

$$f(W) = \{ f(a) : a \in W \} \quad (\text{onto})$$

We say that  $f$  is surjective if

$$f(A) = \text{im } f = B \quad (\text{IE, "f takes every possible value"})$$

$a \in A, f(a) \in B$  elements

$f(A)$  is a set

$A$   
not an  
elt

$f(W)$

$$f(A) \subseteq B$$

"

$$\{ f(a) : a \in A \}$$

To prove  $f(A) = B$ ,  
only need to prove  
 $B \subseteq f(A)$ .





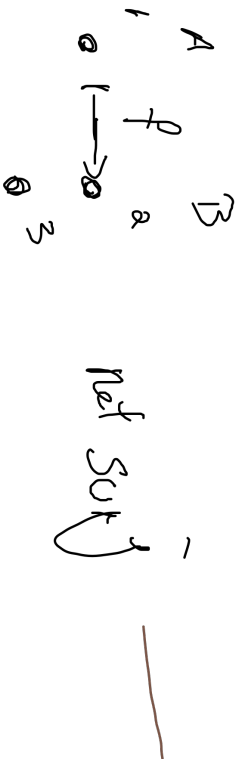
$$f(A) = \{ f(a) : a \in A \}$$

$$= \{ f(a) : a \in \{1, 2, 3\} \}$$

$$= \{ f(1), f(2), f(3) \} = \{ 1, 2, 3 \} = B$$

$$W = \{1, 2\}$$

$$f(W) = \{ f(a) : a \in W \} = \{ f(1), f(2) \} = \{ 1, 2 \} = W$$



$$f(A) = \{ f(a) : a \in A \} = \{ 1, 2 \} \neq \{ 1, 2, 3 \} = B$$

$$B \neq f(A) \text{ b/c } \exists b \in B, b \notin f(A)$$

the defn of  $b \in f(A)$

To prove  $f(A) = B$ , need to show

$$\forall b \in B, \exists a \in A \text{ s.t. } f(a) = b$$

To prove  $f(A) \neq B$ , need to show

$$\exists b \in B \text{ s.t. } \forall a \in A, f(a) \neq b$$

$$\mathbb{R} \xrightarrow{f} \mathbb{R}$$

$$x \mapsto x^2$$

not sur b/c

$-1 \notin \text{im } f$

$$\exists q \in \mathbb{R} \text{ s.t. } f(q) = -1$$

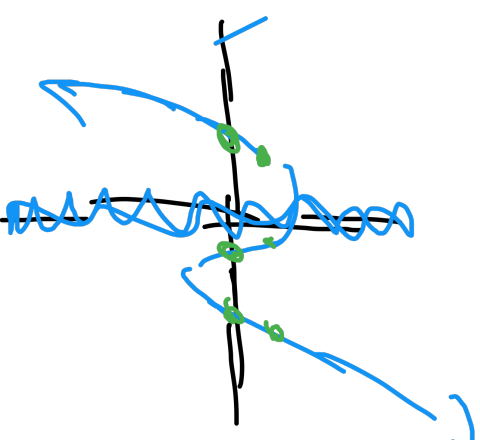
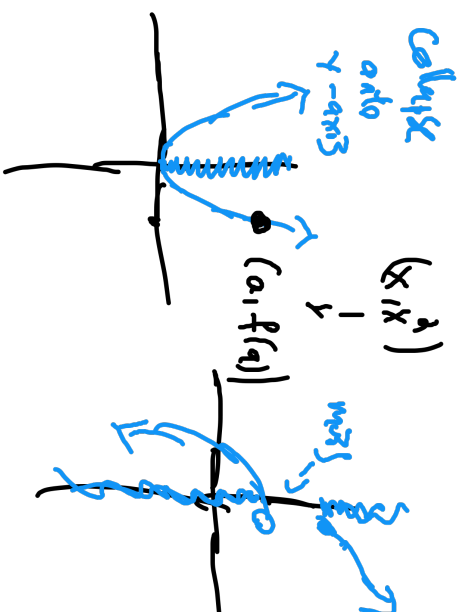
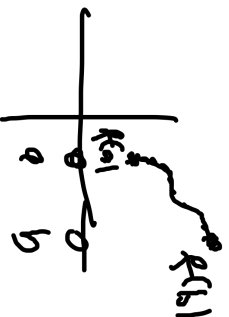
$$\exists q \in \mathbb{R} \text{ s.t. } q^2 = -1$$

$$\text{b/c } q^2 \geq 0 \forall q \in \mathbb{R}$$

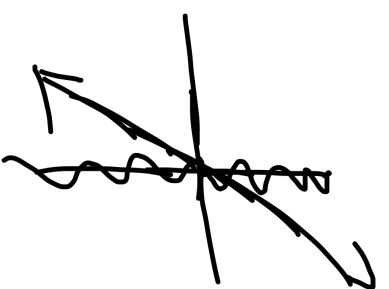
$$f(\mathbb{R}) = \text{im } f = \mathbb{R}_{\geq 0}$$

(b y "continuity")

$$f(0) = 0 \quad \text{AND} \quad \lim_{x \rightarrow \infty} f(x) = \infty$$



$$\lim_{x \rightarrow \pm \infty} f(x) = \pm \infty$$

$$\begin{aligned} \mathbb{R} &\xrightarrow{f} \mathbb{R} \\ x &\mapsto 2x+1 \\ \inf &= \mathbb{R} \end{aligned}$$


Pf: I.V.T, use continuity + limits.

Pfa: Claim:  $f(\mathbb{R}) = \mathbb{R}$ .

Automatic:  $f(\mathbb{R}) \subseteq \mathbb{R}$ .

Goal:  $\mathbb{R} \subseteq f(\mathbb{R})$ .

Let  $b \in \mathbb{R}$ . (WTS  $b \in f(\mathbb{R})$ ).

(Need  $\exists a \in \mathbb{R}$  s.t.  $f(a) = b$ , i.e.,  $2a+1=b$ )

$$a = \frac{b-1}{2}$$

$$\text{Let } a = \frac{b-1}{2}. \text{ Then } a \in \mathbb{R} \text{ and } f(a) = 2\left(\frac{b-1}{2}\right) + 1 = b-1+1 = b.$$

Thus  $b \in f(\mathbb{R})$